# Pattern Recognition Lab

## Experiment No 2

## "Implementing the Perceptron algorithm for finding the weights of a Linear Discriminant function."

You are given the following sample points in a 2-class problem:

$\omega_1$ = (1, 1), (1, -1), (4, 5)

$\omega_2$ = (2, 2.5), (0, 2), (2, 3)

1. Plot all sample points from both classes, but samples from the same class should have the same color and marker. Observer if these two classes can be separated with a linear boundary.
2. Consider the case of a second order polynomial discriminant function. Generate the high dimensional sample points y, as discussed in the class. We used the following formula:

   $$y = [x_1^2 \quad x_2^2 \quad x_1 * x_2 \quad x_1 \quad x_2 \quad 1]$$

   Also, normalize any one of the two classes.
3. Use Perceptron Algorithm (one at a time) for finding the weight-coefficients of the discriminant function (i.e., values of w) boundary for your linear classifier in question 2. Here $\alpha$ is the learning rate and $0 < \alpha \leq 1$.

$$\underline{\mathbf{w}}(\mathbf{i+1}) = \underline{\mathbf{w}}(\mathbf{i}) + \alpha\, \underline{\tilde{y}}_m^{(k)} \qquad \textbf{if } \underline{\mathbf{w}}^{\mathbf{T}}(\mathbf{i})\ \underline{\tilde{y}}_m^{(k)} \leq 0$$

$$\textbf{(i.e., if } \underline{\tilde{y}}_m^{(k)} \textbf{ is misclassified)}$$

$$= \underline{\mathbf{w}}(\mathbf{i}) \qquad \textbf{if } \underline{\mathbf{w}}^{\mathbf{T}}(\mathbf{i})\ \underline{\tilde{y}}_m^{(k)} > 0$$

4. Draw the decision boundary between the two classes using the values of w and the following MATLAB codes.

**Note:** Useful MATLAB Functions: **sym, solve, subs**

```
% Plotting the discriminant function and the prototypes.
syms x1 x2;
s=sym(10*x1*x1-6*x2*x2+24*x1*x2-24*x1-68*x2+65);
s2=solve(s,x2);
```

```
xvals1=[-10:0.01:10];
xvals2(1,:)=subs(s2(1),x1,xvals1);
plot(xvals1,xvals2(1,:),'k');
grid, hold,
%Class S1
plot(1,1,'ro');
plot(1, -1,'ro');
plot(4,5,'ro');
%Class S2
plot(2,2,'gs');
plot(0,2,'gs');
plot(2,3,'gs');
```